Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

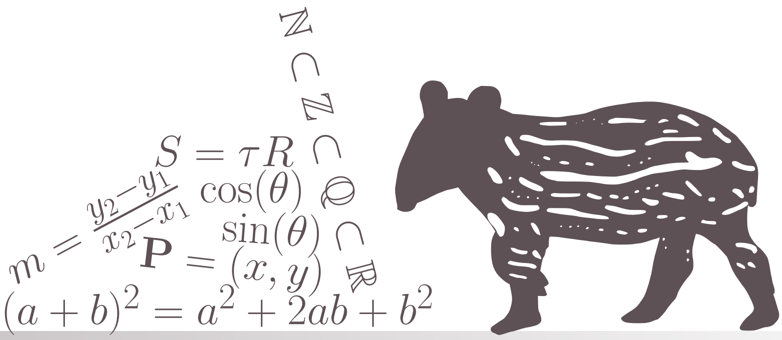
Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1 MATHEMATICAL BACKGROUND



This is a test text for the **Mathematical background** chapter.

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It should be!

1.1 PREFACE

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Testing some equations: let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\boldsymbol{c} = \alpha \boldsymbol{x} + \beta \boldsymbol{y} \in \mathbb{R}^n. \tag{1.1}$$

However, if $\boldsymbol{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha \boldsymbol{x} \in \mathbb{R}^m$, and therefore $\boldsymbol{c}$ is not defined (i.e. Equation 1.1 is irrelevant).

Now, this is a test of the *aligned* environment, which allows for multiple (correctly-aligned) lines of equations and having the label only once (instead of per line, as in the *align* environment):

$$\begin{aligned}
\sin(2x) &= \frac{1}{2\mathrm{i}}\left(\mathrm{e}^{2x\mathrm{i}} - \mathrm{e}^{-2x\mathrm{i}}\right) \\
&= \frac{1}{2\mathrm{i}}\left(\left[\mathrm{e}^{x\mathrm{i}}\right]^2 - \left[\mathrm{e}^{-x\mathrm{i}}\right]^2\right) \\
&= \underbrace{\frac{1}{2\mathrm{i}}\left(\mathrm{e}^{x\mathrm{i}} - \mathrm{e}^{x\mathrm{i}}\right)}_{\sin(x)} \underbrace{\left(\mathrm{e}^{x\mathrm{i}} + \mathrm{e}^{x\mathrm{i}}\right)}_{2\cos(x)} \\
&= 2\sin(x)\cos(x).
\end{aligned} \tag{1.2}$$

Note 1.1 Test note

This is a test note. I will try to also refernce it later.

Now, I wish to refernce Note 1.1. Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = \begin{bmatrix} 1, & 2, & -3 \end{bmatrix}$ and $y = \begin{bmatrix} 5, & -1 \end{bmatrix}$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha \boldsymbol{x} \in \mathbb{R}^3$, while $\beta \boldsymbol{y} \in \mathbb{R}^2$. This means that $\alpha \boldsymbol{x} + \beta \boldsymbol{y}$ is not defined.

...let's check if referencing an example works correctly: can you see Example 1.1?

I believe it is now time for some figures! See Figure 1.1.

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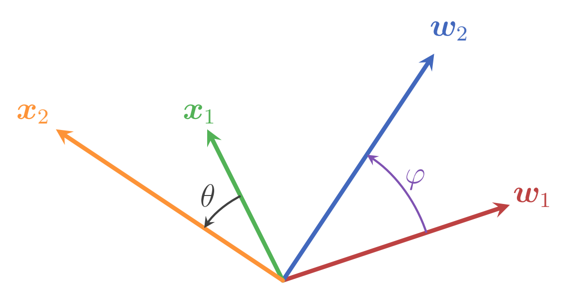

Figure 1.1: This is the first test figure. Does it look ok?

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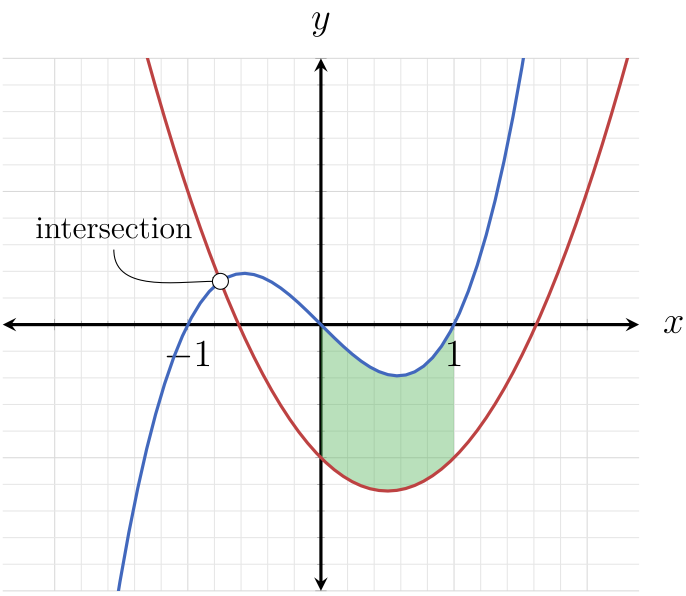

Figure 1.2: And this is another test figure! I hope it works.

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$$T\left(\alpha\vec{x}+\beta\vec{y}\right)=\alpha T(\vec{x})+\beta T(\vec{y})$$

$$\langle\vec{a}\mid\vec{b}\rangle=\vec{a}\cdot\vec{b}=\sum_{i=1}^{n}a_ib_i$$

$$\vec{a}\cdot\vec{b}=0\Leftrightarrow\vec{a}\perp\vec{b}$$

$$\mathbb{I}_n=\begin{pmatrix}1&0&\dots&0\\0&1&\dots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\dots&1\end{pmatrix}$$

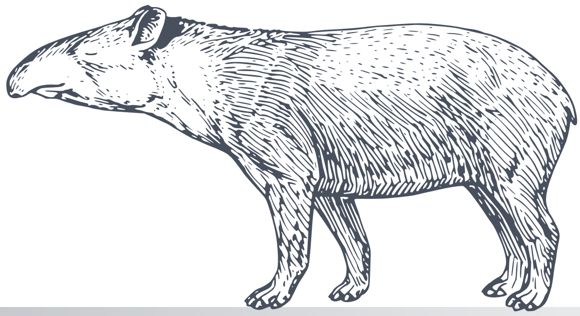
$$(\mathbf{A}\cdot\mathbf{B})^\intercal=\mathbf{B}^\intercal\cdot\mathbf{A}^\intercal$$

$$\vec{v}=\alpha_1\hat{e}_1+\alpha_2\hat{e}_2+\dots+\alpha_n\hat{e}_n$$

$$\mathbf{A}\vec{x}=\lambda\vec{x}$$

$$\det\left(\mathbf{A}\cdot\mathbf{B}\right)=\det(\mathbf{A})\cdot\det(\mathbf{B})$$

$$\mathbf{C}=\mathbf{A}\cdot\mathbf{B}\Rightarrow c^i_j=\langle a^i\mid b_j\rangle$$



CHAPTER 2 THE BASICS OF LINEAR ALGEBRA

This chapter introduces the most important concepts of linear algebra, such as vector spaces, linear transformations, and eigenvectors and eigenvalues. Starting with simple 2- and 3-dimensional geometric concepts, the chapter guides the reader through the topics by introducing them in a visual and intuituve way, only generalizing them afterwards.

After reading this chapter and successfully completing the assignments at the end, the reader should have an intuitive understanding of basic linear algebra.

2.1 VECTORS AND VECTOR SPACES

2.2 LINEAR TRANSFORMATIONS

(HERE COMES LOTS OF INTRO TEXT ON LINEAR TRANSFORMATIONS WHICH I WILL WRITE LATER)

Linear transformations have a very nice property which makes applying them on different vectors extremely efficient. Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ which is a composition of scaling by a factor of 2 in the y-diretion followed by a rotation of 90° anti-clockwise around the origin. Now let's say that we wish to calculate how T transforms the vector

$$\boldsymbol{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

In principle, we could simply apply the operations one after the other on $\boldsymbol{a}$: scaling by 2 in the y-direction yields the vector

$$\tilde{\boldsymbol{a}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

and then rotating it yields the final vector

$$T(\boldsymbol{a}) = \begin{bmatrix} -2 \\ 3 \end{bmatrix}. \tag{2.1}$$

Now, what happens if we wish to calculate a more complicated linear transformation, which is both composed out of many more basic transformations - and also acts on higher-dimensional vectors? Performing each transformation step might not be so straight-forward, and is definitely not an efficient process. In fact, even if we're dealing with a simple example just like the one we just saw, we would still need to apply it every time we want to calculate how a new vector is transformed.

It turns out that thanks to the simplicity of linear transformations [1] we only need to perform a few simple calculations per transformation to practically get any possible application of it on any vector!

[1] more correctly: to the strict rules they enforce

To see how this is the case, let's go back to our simple 2-dimensional example. First, recall that we can express any vector in terms of the standard basis vectors (REF), which in this case gives

$$\boldsymbol{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

. We can then apply the defining properties of linear transformations (REF) on $\boldsymbol{a}$:

$$\begin{aligned} T \begin{bmatrix} 3 \\ 1 \end{bmatrix} &= T \left(3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= T \left(3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + T \left(1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= 3T \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1T \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= 3T\left(\boldsymbol{e}_1\right) + T\left(\boldsymbol{e}_2\right). \end{aligned} \tag{2.2}$$

Now, applying T to the two standard basis vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ we get

$$\begin{aligned} T\left(\boldsymbol{e}_1\right) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ T\left(\boldsymbol{e}_2\right) &= \begin{bmatrix} -2 \\ 0 \end{bmatrix}, \end{aligned} \tag{2.3}$$

and then

$$\begin{aligned} 3T\left(\boldsymbol{e}_1\right) + T\left(\boldsymbol{e}_2\right) &= 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \end{aligned}$$

which is exactly what we got when we calculated T's effect directly on $\boldsymbol{a}$ (cf. Equation 2.1).

Notice exactly what happened: in the standard basis representation, $\boldsymbol{a}$ had the coefficients 3 and 1 for $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, respectively. When applying T on $\boldsymbol{a}$ these two coefficients remained, but the basis changed: instead of $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$, the two basis vectors are $\big\{T(\boldsymbol{e}_1), T(\boldsymbol{e}_2)\big\}$.

Of course, this is not unique to $\boldsymbol{a}$. For example, let's look at another vector,

$$\boldsymbol{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

Its coefficients in the standard basis representation are 2 and -4, respectively. So we expect $T(\boldsymbol{b})$ to be

$$\begin{aligned} T(\boldsymbol{b}) &= 2T(\boldsymbol{e}_1) - 4T(\boldsymbol{e}_2) \\ &= 2\begin{bmatrix} 0 \\ 1 \end{bmatrix} - 4\begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \end{aligned} \tag{2.4}$$

If we now calculate $T(\boldsymbol{b})$ directly, we get that scaling $\boldsymbol{b}$ by 2 in the y-direction gives the vector

$$\tilde{\boldsymbol{b}} = \begin{bmatrix} 2 \\ -8 \end{bmatrix},$$

and rotating this vector by $90°$ anti-clockwise gives

$$T(\boldsymbol{b}) = \begin{bmatrix} 8 \\ 2 \end{bmatrix},$$

exactly as we got in Equation 2.4.

What this insight shows us is that given a linear transformation, we only need to know how it transforms the respective standard basis vectors in order to calculate how it transform any vector in its domain. The transformed version of the vector would simply be the linear combination of these transformed basis vectors, with the coefficients given by the vector's presentation in the standard basis. It really can't get any simpler than that!

To make this a bit more formal, let's look at the most generic case: say $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation [2]. Then applying it to a vector $\boldsymbol{v} \in \mathbb{R}^n$ in its standard basis representation

$$\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + \cdots + v_n \boldsymbol{e}_n$$

we get

$$\begin{aligned} T(\boldsymbol{v}) &= T \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= T \left(v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + \ldots v_n \boldsymbol{e}_n \right) \\ &= v_1 T(\boldsymbol{e}_1) + v_2 T(\boldsymbol{e}_2) + \cdots + v_n \boldsymbol{e}_n . \end{aligned} \tag{2.5}$$

This is an incredibly powerful insight which allows us to save a lot of calculation - and which we will use in the next section to make the calculations of linear transformations *even more* concise and simple [3].

[2] we'll deal with the more general case of $T : \mathbb{R}^n \to \mathbb{R}^m$ where $n \neq m$ in the next section

[3] **spoiler**: matrices!

2.3 MATRICES

2.4 LINEAR EQUATIONS

2.5 EIGENVECTORS AND EIGENVALUES

2.6 MATRIX DECOMPOSITIONS

$$T\left(\alpha\vec{x}+\beta\vec{y}\right)=\alpha T(\vec{x})+\beta T(\vec{y})$$

$$\langle\vec{a}\mid\vec{b}\rangle=\vec{a}\cdot\vec{b}=\sum_{i=1}^{n}a_ib_i$$

$$\vec{a}\cdot\vec{b}=0\Leftrightarrow\vec{a}\perp\vec{b}$$

$$\mathbb{I}_n=\begin{pmatrix}1&0&\dots&0\\0&1&\dots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\dots&1\end{pmatrix}$$

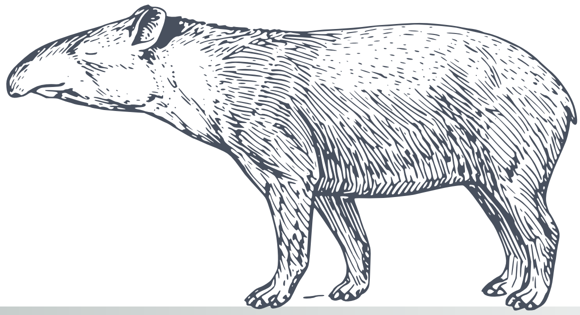
$$(\mathbf{A}\cdot\mathbf{B})^{\intercal}=\mathbf{B}^{\intercal}\cdot\mathbf{A}^{\intercal}$$

$$\mathbf{A}\vec{x}=\lambda\vec{x}$$

$$\vec{v}=\alpha_1\hat{e}_1+\alpha_2\hat{e}_2+\dots+\alpha_n\hat{e}_n$$

$$\det\left(\mathbf{A}\cdot\mathbf{B}\right)=\det(\mathbf{A})\cdot\det(\mathbf{B})$$

$$\mathbf{C}=\mathbf{A}\cdot\mathbf{B}\Rightarrow c_{ij}=a_i\cdot b_j$$



CHAPTER 3

INTRODUCING SOME FORMALISM

This is a test text for the Mathematical background chapter. Is this what's written in the actuall chapter header? It should be!

$$T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} \mid \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

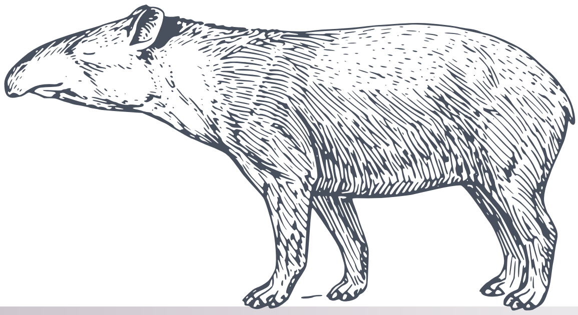
$$(\mathbf{A} \cdot \mathbf{B})^{\intercal} = \mathbf{B}^{\intercal} \cdot \mathbf{A}^{\intercal}$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER 4
ADVANCED TOPICS IN LINEAR ALGEBRA

This is a test text for the Mathematical background chapter. Is this what's written in the actuall chapter header? It should be!

$$T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} \mid \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

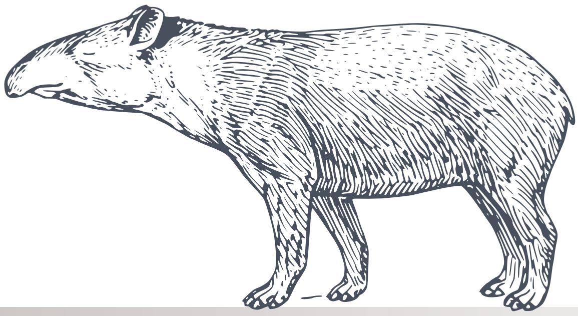
$$(\mathbf{A} \cdot \mathbf{B})^\intercal = \mathbf{B}^\intercal \cdot \mathbf{A}^\intercal$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER 5

FURTHER RELATED TOPICS

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